

Customer Churn Classification Model Comparison						
Validation and Final Test Results						
Model	Split	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Gradient Boosting Classifier	Validation	0.7616	0.5351	0.7865	0.6369	0.8376
Bayesian Ensemble	Validation	0.7701	0.5503	0.7402	0.6313	0.8374
Average Ensemble	Validation	0.7569	0.5302	0.7509	0.6215	0.8373
Random Forest Classifier	Validation	0.7616	0.5381	0.7295	0.6193	0.8329
Logistic Regression	Validation	0.7531	0.5237	0.7865	0.6287	0.8319
Decision Tree Classifier	Validation	0.7408	0.5085	0.7473	0.6052	0.8144
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Support Vector Classifier	Validation	0.7417	0.5097	0.7473	0.6061	0.8069
K-Nearest Neighbors Classifier	Validation	0.6859	0.4452	0.7367	0.5550	0.7769
Bayesian Ensemble	Test	0.7711	0.5470	0.7893	0.6462	0.8531
Gradient Boosting Classifier	Test	0.7512	0.5191	0.8250	0.6372	0.8517
Random Forest Classifier	Test	0.7644	0.5385	0.7750	0.6354	0.8482
Logistic Regression	Test	0.7294	0.4934	0.8036	0.6114	0.8482
Decision Tree Classifier	Test	0.7502	0.5192	0.7714	0.6207	0.8307
Support Vector Classifier	Test	0.7446	0.5116	0.7857	0.6197	0.8214
K-Nearest Neighbors Classifier	Test	0.6944	0.4544	0.7643	0.5699	0.7860

Note: The models were selected and compared using the validation dataset. The test dataset remained untouched until the final evaluation. The Bayesian Ensemble produced the highest final test ROC-AUC score of 0.8531.